



Juki Corp's RX-8 (Credit: <https://www.juki.co.jp>)

time, and other adjustments along the entire assembly process.

The industry is working towards faster machines every year to match the increased speed demand. Yamaha Motor's ultra-high-speed modular Z:TA-R YSM40R is one of the recent launches claiming to offer 200,000 CPH. The speed is said to be achieved by edge technology, including innovative high-speed rotary heads and linear motors incorporating new, high-speed algorithms, with the machine width of just one metre.

Juki Corp's RX-8 is another recent launch that claims to have the highest placement rate per square metre with CPH up to 100,000. It is compact and modular, featuring 998mm width. The new P20 placement head, designed for placing ultra-small chips and small integrated circuits (ICs), is credited for high-speed picking and placing from the single reel.

Intelligent camera systems. A camera system with high resolution improves the overall speed of the machine. New machines focus on their camera systems, like the one in the Neoden 8 PnP machine. It comes with the HR Flyvision camera



Vision system inside Neoden 8 PnP machine (Credit: <https://neodenusa.com>)

system for component recognition and Double Mark cameras to reach the extreme end feeders for better calibration.

Component handling. The PnP machines handle the minimum and maximum component dimensions according to the production needs for chip-scale package (CSP), ball-grid array (BGA), and other odd parts. The machines need various alignment methods to handle such a range of components, say, from minimum sizes of 0402, 0201, or 01005. Tiny components smaller than 0402 may require additional nozzles. The machines with single alignment give more precision and speed. An alternative alignment system can handle large component sizes.

However, one should remember, not all machines that can handle large components can also inspect them. Hence, most of the new launches with the latest vision system and software can take on

everything from 01005-size flip chips and CSPs to large quad flat packages (QFPs) and ball grid arrays (BGAs). Even tall, odd-shaped and through-hole components can be mounted with ease.

For instance, ASM Assembly Systems has a complete solution from component pickup, image recognition to placement of odd-shaped components. Its Siplace TX models are said to work with up to 48,000 CPH with minimum 0201 (metric) and maximum component size of 8.2mm × 8.2mm × 4mm.

Number of pick up heads. A PnP machine has single or multi-heads to grab different shapes and heavy parts. Equipped with a number of nozzles to grasp the components, the head is movable in different directions simultaneously to ensure high productivity and flexibility.

For instance, the Yamaha YRM20 comes with a rotary RM head that is claimed to achieve 115,000 CPH. It features a one-head solution that can carry components of odd shapes and eliminates the need for head replacements, thus improving the component feed speed.

Similarly, the Neoden 8 model has eight synchronised nozzle heads to give repeatable placement accuracy.

Smart feeders. Recent machines focus also on the feeder capacity.



ASM Assembly Systems machine (Credit: www.asm-smt-tools.com)